

## $\theta$ 6-dimensional biophysical vector

substitution chemistry

- $\Delta$  hydrophobicity
- $\Delta$  side-chain volume
- $\Delta$  charge

structural context

- fold-core location
- functional-switch residue

annotation

- hotspot / pathogenic

TP53



R175H

R  $\rightarrow$  H



shared  
feature  
space

alternative feature spaces tested later: ESM, ESM +  $\theta$